Titration Calculations

Answer key with worked solutions

A. Titres

1. Definitions.

> **Titre**: volume delivered from the burette. **Aliquot**: an accurately measured portion of a solution (by pipette). **End point**: where the indicator changes colour. **Equivalence point**: where exactly the stoichiometric amount has been added. **Concordant titres**: accurate titres that agree closely (within 0.10 mL).

2. Hydrochloric acid.

> (a) Mean of the concordant titres (rough left out):
>
> $$\bar{V} = \frac{(22.60 + 22.70 + 22.65)\ \text{mL}}{3} = \mathbf{22.65\ mL}$$
>
> Deviations −0.05, +0.05, 0.00 mL: $s = \sqrt{\dfrac{0.0050\ \text{mL}^2}{2}} = \mathbf{0.05\ mL}$
>
> (b) $n(\text{NaOH}) = 0.1050\ \text{mol/L} \times 0.02265\ \text{L} = 2.378 \times 10^{-3}\ \text{mol} = n(\text{HCl})$
>
> $c(\text{HCl}) = \dfrac{2.378 \times 10^{-3}\ \text{mol}}{0.02500\ \text{L}} = \mathbf{0.09513\ mol/L}$

B. Standardisation

3. Sodium carbonate.

> (a) $n = \dfrac{1.325\ \text{g}}{105.99\ \text{g/mol}} = 0.01250\ \text{mol}$; $c = \dfrac{0.01250\ \text{mol}}{0.2500\ \text{L}} = \mathbf{0.05000\ mol/L}$
>
> (b) Aliquot: $0.05000\ \text{mol/L} \times 0.02500\ \text{L} = 1.250 \times 10^{-3}\ \text{mol}\ \text{Na}_2\text{CO}_3$; $\text{HCl} = 2 \times 1.250 \times 10^{-3} = 2.500 \times 10^{-3}\ \text{mol}$
>
> $c(\text{HCl}) = \dfrac{2.500 \times 10^{-3}\ \text{mol}}{0.02480\ \text{L}} = \mathbf{0.1008\ mol/L}$

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4. KHP.

> $n = \dfrac{0.5105\ \text{g}}{204.22\ \text{g/mol}} = 2.500 \times 10^{-3}\ \text{mol} = n(\text{NaOH});\ c = \dfrac{2.500 \times 10^{-3}\ \text{mol}}{0.02495\ \text{L}} =$ **$0.1002\ \text{mol/L}$**

C. Diprotic acids and dilutions

5. Sulfuric acid.

> $n(\text{NaOH}) = 0.2000\ \text{mol/L} \times 0.01825\ \text{L} = 3.650 \times 10^{-3}\ \text{mol};\ n(\text{H}_2\text{SO}_4) = \frac{1}{2} \times 3.650 \times 10^{-3} = 1.825 \times 10^{-3}\ \text{mol};\ c = \dfrac{1.825 \times 10^{-3}\ \text{mol}}{0.02500\ \text{L}} =$ **$0.07300\ \text{mol/L}$**

6. Vinegar.

> Aliquot: $0.1000\ \text{mol/L} \times 0.02330\ \text{L} = 2.330 \times 10^{-3}\ \text{mol}$
> Flask: $\dfrac{250.0\ \text{mL}}{25.00\ \text{mL}} = 10$ aliquots, so 2.330×10^{-2} mol, all from the 20.00 mL of vinegar.
> $c = \dfrac{2.330 \times 10^{-2}\ \text{mol}}{0.02000\ \text{L}} =$ **$1.165\ \text{mol/L}$**

D. Back titration

7. Antacid.

> HCl added: $0.5000\ \text{mol/L} \times 0.02500\ \text{L} = 1.250 \times 10^{-2}\ \text{mol}$
> HCl left over (= NaOH): $0.2500\ \text{mol/L} \times 0.01120\ \text{L} = 2.800 \times 10^{-3}\ \text{mol}$
> HCl reacted: $1.250 \times 10^{-2} - 2.800 \times 10^{-3} = 9.700 \times 10^{-3}\ \text{mol}$
> Mg(OH)_2 (1 : 2): $\frac{1}{2} \times 9.700 \times 10^{-3} = 4.850 \times 10^{-3}\ \text{mol}$; mass $= 4.850 \times 10^{-3}\ \text{mol} \times 58.32\ \text{g/mol} = 0.2829\ \text{g}$
> Percent: $\dfrac{0.2829\ \text{g}}{0.300\ \text{g}} \times 100\% =$ **94.3%**

E. Challenge

8. Uncertainty.

> Two readings: $2 \times 0.05\ \text{mL} = 0.10\ \text{mL}$; $\dfrac{0.10\ \text{mL}}{22.65\ \text{mL}} \times 100\% =$ **0.44%**. To reduce it, make the titre **larger** (for example, use a more dilute titrant or a larger aliquot), so the same ±0.10 mL is a smaller fraction of the titre.

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